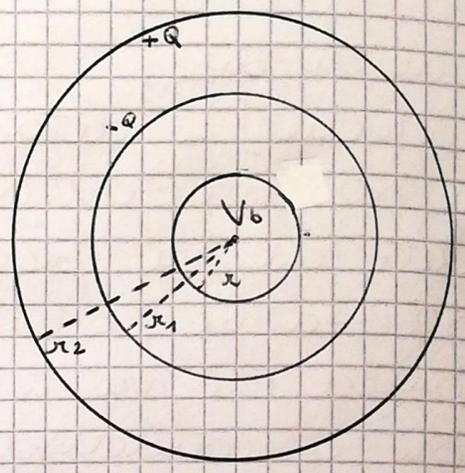
APPELLO 14/01/2019

ESERCIZIO 1

Considero $V_\infty = 0$.

$Q_0 = V_0 \varepsilon_0 4\pi R$



SE $0<r<R$

$\underline{E} = 0$

SE $R<r<R_1$

$\underline{E} = \frac{V_0 R}{r^2} \underline{u}_r$

SE $R_1<r<R_2$

$E = 0$

SE $r>R_2$

$E = \frac{V_0 R}{r^2} \underline{u}_r$

POTENZIALE

SE $r>R_2$

$V(r) - \overset{0}{V_\infty} = -\int_\infty^r \frac{V_0 R}{r'^2} dr' = \frac{V_0 R}{r}$

SE $R_1<r<R_2$

$V(r) - V(R_2) = 0 \Rightarrow V(r) = V(R_2) = \frac{V_0 R}{R_2}$

SE $R<r<R_1$

$V(r) - V(R_1) = -\int_{R_1}^r \frac{V_0 R}{r^2} dr = V_0 R \left(\frac{1}{r} - \frac{1}{R_1}\right)$

$V(r) = V_0 R \left(\frac{1}{r} - \frac{1}{R_1} + \frac{1}{R_2}\right)$

$V(R) = V_0 R \left(\frac{1}{R} - \frac{1}{R_1} + \frac{1}{R_2}\right)$

se $V(R_1) = 0$

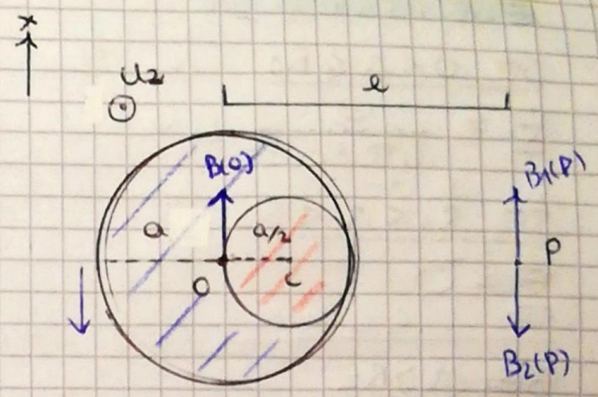
$R<r<R_1$

$V(r) - V(R_1) = -\int_{R_1}^r \frac{V_0 R}{r^2} \Rightarrow V(r) = V_0 R \left(\frac{1}{r} - \frac{1}{R_1}\right)$

$\Rightarrow V_2 = V(R) = V_0 R \left(\frac{1}{R} - \frac{1}{R_1}\right)$

ESERCIZIO 2

$$\underline{J} = \frac{I}{\pi a^2}\,\underline{u}_z$$



Per il filo pieno si ha:

SE $r<a$

$$2\pi r B = \frac{\mu_0}{2} J \pi r^2$$

$$\underline{B}_1 = \frac{\mu_0 I r}{2\pi a^2}\,\underline{u}_t$$

SE $r>a$

$$\underline{B}_2 = \frac{\mu_0 I}{2\pi r}\,\underline{u}_t$$

Per la cavità:

$$\underline{B}_2 = \frac{\mu_0 I r'}{2\pi a^2}\,\underline{u}_t \qquad B_2 = \frac{\mu_0 I}{8\pi r'}\,\underline{u}_t$$

Il campo totale in O è

$$\underline{B}_{TOT}(O) = 0 - \left(-\frac{\mu_0 I}{4\pi a}\right)\underline{u}_x = \frac{\mu_0 I}{4\pi a}\,\underline{u}_x$$

In C invece

$$\underline{B}_{TOT}(C) = \left(\frac{\mu_0 I}{4\pi a} - 0\right)\underline{u}_x = \frac{\mu_0 I}{4\pi a}\,\underline{u}_x$$

e infine

$$\underline{B}_{TOT}(P) = \left(\frac{\mu_0 I}{2\pi e} - \frac{\mu_0 I}{8\pi(e-a/2)}\right)\underline{u}_x$$

ESERCIZIO 4